# Year 13 Further Maths - Elastic strings

Similar questions - Circular motion and projectiles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

**1** A right circular cone $C$ of height 3 m and base radius 4 m has its base fixed to a horizontal plane. One end of a light elastic string of natural length 1.5 m and modulus 50 N is fixed to the vertex of $C$. The other end is attached to a particle $P$ of mass 3 kg. $P$ moves in a horizontal circle with constant speed in contact with the smooth curved surface of $C$. The extension of the string is 1 m.

**(a)** Find the tension in the string. **[2]**

**(b)** Find the speed of $P$. **[7]**

**2** One end of a light elastic string of natural length 0.8 m and modulus $\lambda$ N is attached to a fixed point $O$ on a smooth horizontal table. The other end is attached to a particle $A$ of mass 0.5 kg which moves on the table in a circle with constant speed $v\ \mathrm{m\,s^{-1}}$. The extension of the string is $x$ m.

**(a)** Show that $\lambda x^2 + 0.8\lambda x - 0.4v^2 = 0$. **[3]**

**(b)** For $\lambda = 120$ and $v = 4$, find $x$. **[2]**

**(c)** Show that for large $\lambda$, $\lambda x \approx mv^2$ and explain what this means for an inextensible string. **[3]**

**3** A trainee of mass 80 kg stands on the edge of a platform. A light inextensible cord of length 4 m joins the trainee to a hook $H$ at the same level as the platform and 4 m horizontally away, with the cord straight and taut. The trainee is projected downwards with speed $5\ \mathrm{m\,s^{-1}}$ perpendicular to the cord and swings in a vertical circle. The ground is 6 m below the level of $H$. Model the trainee as a particle and ignore air resistance.

**(a)** The cord is released from $H$ when the speed of the trainee is $8\ \mathrm{m\,s^{-1}}$. Show that the tension just before release is 1670 N (3 s.f.). **[6]**

**(b)** Find the speed and the angle to the horizontal at which the trainee reaches the ground. **[7]**

**Question 1**

(a) $T = \dfrac{50 \times 1}{1.5} = 33.3$ N

(b) $P$ is 2.5 m from the vertex along the slope; semi-vertical angle $\alpha$ has $\sin\alpha = \frac{4}{5}$, $\cos\alpha = \frac{3}{5}$, so the radius is $r = 2.5 \times 0.8 = 2$ m. Vertically: $T\cos\alpha + R\sin\alpha = 3g$: $20 + 0.8R = 29.4$, $R = 11.75$ N. Horizontally (inwards): $T\sin\alpha - R\cos\alpha = \dfrac{3v^2}{2}$: $26.67 - 7.05 = 1.5v^2$, $v = 3.62\ \mathrm{m\,s^{-1}}$

**Question 2**

(a) $T = \dfrac{\lambda x}{0.8}$ provides the centripetal force at radius $0.8 + x$: $\dfrac{\lambda x}{0.8} = \dfrac{0.5v^2}{0.8+x}$, so $\lambda x(0.8+x) = 0.4v^2$

(b) $120x^2 + 96x - 6.4 = 0$, $x = 0.0619$ m

(c) $\lambda x^2 + \lambda l x = lmv^2$ gives $x = \frac{l}{2}\left(-1 + \sqrt{1 + \frac{4mv^2}{\lambda l}}\right) \approx \frac{l}{2} \cdot \frac{2mv^2}{\lambda l} = \frac{mv^2}{\lambda}$, so $\lambda x \approx mv^2$, i.e. the tension is $mv^2/l$ (the radius is effectively $l$): exactly the tension in an inextensible string of length $l$

**Question 3**

(a) Height fallen $d = \dfrac{8^2 - 5^2}{2g} = 1.99$ m, so the cord makes angle $\phi$ below the horizontal with $\sin\phi = \frac{1.99}{4} = 0.497$.

Radially towards $H$: $T - mg\sin\phi = \dfrac{mv^2}{4}$, $T = 80\,(16 + 9.8(0.497)) = 1670$ N

(b) Velocity is perpendicular to the cord: $v_x = 8\sin\phi = 3.98$, downward $8\cos\phi = 6.94$. Remaining fall $6 - 1.99 = 4.01$ m: $4.01 = 6.94t + 4.9t^2$, $t = 0.441$ s. Final downward speed $6.94 + 9.8(0.441) = 11.26$; speed $= \sqrt{3.98^2 + 11.26^2} = 11.9\ \mathrm{m\,s^{-1}}$ at $70.5^\circ$ below the horizontal